Type MS · Using vector ratios to compare areas

Answer 1 of 8 · Question 10 part c · 3 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

10 (a) (i)	$\mathbf{a}+\mathbf{c}$	B1
(ii)	$\frac{1}{2}(\mathbf{c}-\mathbf{a})$	B1 (2)
(b)	$\overrightarrow{OX} = OA + AM + \lambda MN$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a})$	M1 A1
	$\mu(\mathbf{a}+\mathbf{c})$	B1
	$\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a}\right)=\mu(\mathbf{a}+\mathbf{c})$	M1
	$1-\frac{1}{2}\lambda=\mu \quad \frac{1}{2}+\frac{1}{2}\lambda=\mu$	M1
	$1-\frac{1}{2}\lambda=\frac{1}{2}+\frac{1}{2}\lambda$	M1
	$\lambda=\frac{1}{2} \quad \mu=\frac{3}{4}$	A1 A1 (8)
(c)	Triangle $XBN=\frac{1}{8}$ of $\frac{1}{2}$ the parallelogram Quadrilateral $OXNC=\frac{7}{8}$ of $\frac{1}{2}$ the parallelogram So Quadrilateral $OXNC=\frac{7}{16}$ of the parallelogram $\therefore$ 7 : 16	M1 M1 A1 (3) **[13]**

Part	Mark	Additional Guidance
(a)(i)	B1	For the correct vector $\mathbf{a}+\mathbf{c}$
(ii)	B1	For the correct vector $\frac{1}{2}(\mathbf{c}-\mathbf{a})$
(b)	M1	For the correct vector statement $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AM} + \lambda\overrightarrow{MN}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ or $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{\mathbf{c}}{2}-\frac{\mathbf{a}}{2}\right)$
	B1ft	For $\overrightarrow{OX}=\mu(\mathbf{a}+\mathbf{c})$ ft their $\overrightarrow{OB}=$'$\mathbf{a}+\mathbf{c}$'
	M1	For equating their two vector statements for $\overrightarrow{OX}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}+\mathbf{c})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}-\mathbf{c})\Rightarrow\mathbf{a}\left(1-\frac{\lambda}{2}\right)+\mathbf{c}\left(\frac{1}{2}+\frac{\lambda}{2}\right)=\mu\mathbf{a}+\mu\mathbf{c}$ $\Rightarrow\mu=1-\frac{\lambda}{2},\ \mu=\frac{1}{2}+\frac{\lambda}{2}$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $1-\frac{\lambda}{2}=\frac{1}{2}+\frac{\lambda}{2}\Rightarrow\lambda=\ldots\Rightarrow\mu=\ldots$ or $1-\mu=\mu-\frac{1}{2}\Rightarrow\mu=\ldots\Rightarrow\lambda=\ldots$

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	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
	ALT	
	M1	For the correct vector statement $\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OX}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{MX} = -\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c})$
	B1ft	$\overrightarrow{MX} = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ ft their $\overrightarrow{MN} = '\frac{1}{2}(\mathbf{c}-\mathbf{a})'$
	M1	For equating the two vector statements for $\overrightarrow{MX}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a}) \Rightarrow \mathbf{c}\left(-\frac{1}{2}+\mu\right) + \mathbf{a}(\mu-1) = \mathbf{c}\frac{\lambda}{2} - \mathbf{a}\frac{\lambda}{2}$ $\Rightarrow \frac{\lambda}{2} = \mu - \frac{1}{2}$ and $-\frac{\lambda}{2} = \mu - 1$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $\mu - \frac{1}{2} = -(\mu-1) \Rightarrow \mu = \left(\frac{3}{4}\right)$ $\frac{\lambda}{2} = 1 - '\frac{3}{4}' \Rightarrow \lambda = \left(\frac{1}{2}\right)$
	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
(c)	M1	For area of $\Delta XBN = \frac{1}{8}\Delta OBC$ so $\frac{1}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ $\left(\Delta OBC = \frac{1}{2} \times OB \times BC \times \sin\angle XBN \right.$ $\left. \Delta XBN = \frac{1}{2} \times \frac{1}{4} OB \times \frac{1}{2} BC \times \sin\angle XBN = \frac{1}{8}\Delta OBC \right)$
	M1	Therefore Quadrilateral $OXNC = \frac{7}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ ft their fraction from the first M mark provided it is $< \frac{1}{2}$
	A1	Quadrilateral $OXNC = \frac{7}{16}$ of the area of parallelogram $OABC$ so ratio is 7:16

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Question number	Scheme	Marks

Type MS · Using vector ratios to compare areas

Answer 2 of 8 · Question 11 part c · 4 marks

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11(a)	(i) $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$	B1
	(ii) $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \mathbf{a} + \frac{1}{2}\left(-\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$	M1A1
	(iii) $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD} = -\mathbf{b} + \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$	M1A1 (5)
(b)	$\overrightarrow{BE} = k\overrightarrow{BD} = k\left(\frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}\right)$	M1
	$\overrightarrow{BE} = -\mathbf{b} + \overrightarrow{OE} = -\mathbf{b} + \lambda\mathbf{a}$	M1
	$\frac{7}{8}k = 1 \Rightarrow k = \frac{8}{7}$	M1
	$\frac{k}{2} = \lambda \Rightarrow \lambda = \frac{4}{7}$	A1 (4)
(c)	$\triangle OAC = \frac{1}{4}\triangle OAB$	M1
	$\triangle OEB = "\frac{4}{7}"\triangle OAB$	A1 ft
	$\frac{\triangle OAC}{\triangle OEB} = \frac{1}{4} \times "\frac{7}{4}" = \frac{7}{16}$	M1
	$\mu = \frac{7}{16}$	A1 (4)
		[13]

Part	Mark	Notes
(a)(i)	B1	For the correct vector $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$
(ii)	M1	For the correct vector statement $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC}$ or $\overrightarrow{OD} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CA}$
	A1	For the correct **simplified** vector $\overrightarrow{OD} = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$
(iii)	M1	For the correct vector statement $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD}$ or $\overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD}$
	A1	For the correct **simplified** vector $\overrightarrow{BD} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$

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Answer 2 of 8 · Question 11 part c · 4 marks

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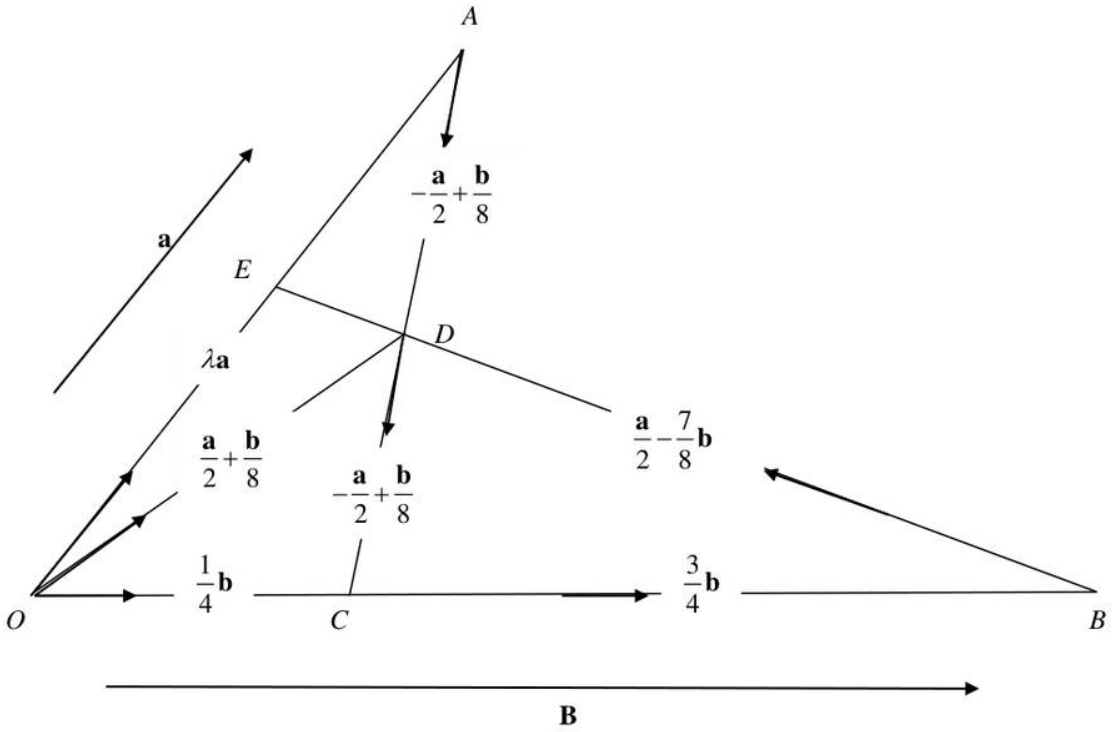
In parts (b) and (c) you must follow through the vectors they have found in part (a)		
(b)	M1	States or implies two paths that will lead to a solution. For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}$ and $\overrightarrow{BE}=\overrightarrow{BO}+\overrightarrow{OE}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\overrightarrow{OD}+\overrightarrow{DE}$ **This is a B mark in Epen**
	M1	For writing their paths as vectors in terms of **a**, **b** λ **and another constant** (e.g. k or μ) For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}=k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ and $\overrightarrow{BE}=-\mathbf{b}+\lambda\mathbf{a}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ OR $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$
	dM1	Equates coefficients of their **two** expressions and attempts to find the value of λ: In triangle OEB • $k\frac{1}{2}\mathbf{a}-k\frac{7}{8}\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}$ $\Rightarrow k\frac{7}{8}=1\Rightarrow k=\frac{8}{7}$ and $\frac{k}{2}=\lambda\Rightarrow\lambda=...$ or in triangle OED, there are two possible expressions for $\overrightarrow{DE}$ • $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ $\Rightarrow k=\frac{1}{8}$ and $\lambda=\frac{1}{2}+\frac{1}{8}\lambda\Rightarrow\lambda=...$ Or $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ $\Rightarrow\frac{7}{8}k=\frac{1}{8}\Rightarrow k=\frac{1}{7}$ and $\lambda=\frac{1}{2}+\frac{1}{7}k\Rightarrow\lambda...$
	A1	For the correct value of λ $\lambda=\frac{4}{7}$
(c)	M1	For stating that $\triangle OAC=\frac{1}{4}\triangle OAB$ or $4\triangle OAC=\triangle OAB$ **1**
	M1	For stating $\triangle OEB="\lambda"\triangle OAB\Rightarrow\triangle OEB="\frac{4}{7}"\triangle OAB$ **2** **This is an A mark in Epen**
	M1	For dividing **1** by **2** $=\frac{\triangle OAC}{\triangle OEB}=\frac{1}{4}\times"\frac{7}{4}"=...$
	A1	For $\mu=\frac{7}{16}$

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Answer 2 of 8 · Question 11 part c · 4 marks

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USEFUL SKETCH



Type MS · Using vector ratios to compare areas

Answer 3 of 8 · Question 8 part c · 4 marks

2024 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

8(a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DC} + \overrightarrow{CB} = 5\mathbf{a} - 2\mathbf{b} - 3\mathbf{a} = 2\mathbf{a} - 2\mathbf{b}$	B1 [1]
(b)	$\overrightarrow{BX} = k\overrightarrow{BD} = k(3\mathbf{a} + 2\mathbf{b})$ $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow k(3\mathbf{a} + 2\mathbf{b}) = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow 3k\mathbf{a} + 2k\mathbf{b} = (5\lambda - 2)\mathbf{a} + (2 - 2\lambda)\mathbf{b}$ $\Rightarrow 3k = 5\lambda - 2 \quad 2k = 2 - 2\lambda$ $\Rightarrow k = \frac{3}{8} \quad \left[\lambda = \frac{5}{8}\right]$	M1 M1 M1 M1A1 [5]
(c)	$\Delta CXD = \frac{5}{8}\Delta BCD$ $\Delta BCD = \frac{3}{8}ABCD$ $\frac{\Delta CXD}{ABCD} = \frac{\frac{5}{8}}{\frac{8}{3}} = \frac{15}{64}$ Ratio of area of triangle CXD : Trapezium $ABCD$ = 15 : 64	M1 M1 M1 A1 [4]
		Total 10 marks

Type MS · Using vector ratios to compare areas

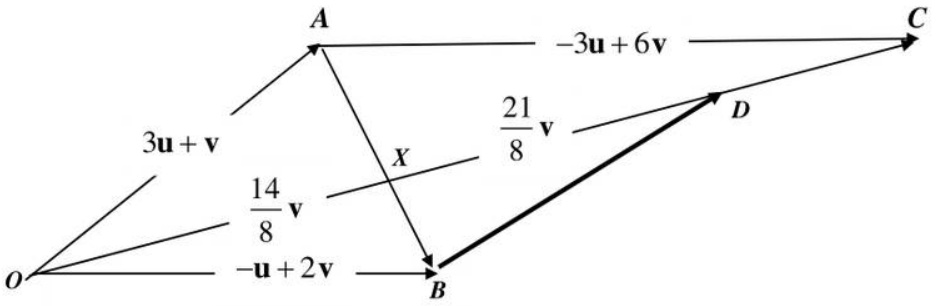
Answer 4 of 8 · Question 8 part c · 5 marks

2025 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

8 (a)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}, = -(3\mathbf{u}+\mathbf{v}) + (2\mathbf{v}-\mathbf{u}) = -4\mathbf{u}+\mathbf{v}$	M1,A1 (2)
(b)	$\overrightarrow{AC} = \lambda(2\mathbf{v}-\mathbf{u})$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = (3\mathbf{u}+\mathbf{v}) + \lambda(2\mathbf{v}-\mathbf{u}) = \mu\mathbf{v}$	M1
	$\mathbf{u}: 0 = 3-\lambda \qquad \mathbf{v}: \mu = 1+2\lambda$	M1
	$\mu = 7 \qquad [\lambda = 3]$	A1 (4)
(c)	$\overrightarrow{OX} = \overrightarrow{OA} + \alpha\overrightarrow{AB} = 3\mathbf{u}+\mathbf{v}+\alpha(-4\mathbf{u}+\mathbf{v})$ or $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AC} + \beta\overrightarrow{CO} = 3\mathbf{u}+\mathbf{v}+3(2\mathbf{v}-\mathbf{u})+\beta(-7\mathbf{v})$	M1
	$\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$	A1
	$\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$	M1
	$\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$	M1
	$\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{35}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{7}{4}\mathbf{v}+\frac{21}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}+\frac{19}{8}\mathbf{v}$	A1 (5)
		Total 11 marks

USEFUL SKETCH

$$\overrightarrow{OD} = \frac{35}{8}\mathbf{v} \quad \text{and} \quad \overrightarrow{AX} = -3\mathbf{u}+\frac{3}{4}\mathbf{v}$$



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Answer 4 of 8 · Question 8 part c · 5 marks

2025 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Use of $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$
	A1	$-4\mathbf{u} + \mathbf{v}$
(b)	M1	For use of $\overrightarrow{AC}$ is parallel to $\overrightarrow{OB}$ to form an expression for $\overrightarrow{AC}$
	M1	For use of their expression for $\overrightarrow{AC}$ to form an expression for $\overrightarrow{OC}$
	M1	For equating for **u** and for **v**
	A1	For $\mu = 7$
(c)		**General principles for marking part (c)** For example: You may see the vector path for $\overrightarrow{BD}$ first - score the 4th M mark • Candidates will need to find vector $\overrightarrow{OX}$ [M1A1] • Candidates will need to use the ratio of bases to find $\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$ **leading** to establish $\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$ [M1] • Check any path they choose on the diagram below. For example, $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OX} + \overrightarrow{XD}$ or [M1] $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OA} + \overrightarrow{AX} + \overrightarrow{XD}$ If a candidate chooses to take a longer path and need to find for example, $\overrightarrow{AX}$ they will need to find a vector. There is no separate mark for this. • For the correct vector
	M1	For an expression for $\overrightarrow{OX}$ including an unknown constant
	A1	For $\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$
	M1	For recognising that triangle OBX and triangle OBD share a base OB and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{OD}$ together with an attempt to find $\overrightarrow{OD}$ OR For recognising that triangle OBX and triangle BXD share a base BX and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{XD}$ together with an attempt to find $\overrightarrow{XD}$
	M1	For method to find $\overrightarrow{BD}$ Award this for any valid path given
	A1	$\overrightarrow{BD} = \mathbf{u} + \frac{19}{8}\mathbf{v}$

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10(a)	(i) $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	B1
	(ii) $\overrightarrow{MY} = \overrightarrow{MA} + \frac{3}{4}\left(\overrightarrow{AB}\right) = \mathbf{a} + \frac{3}{4}(-2\mathbf{a} + 4\mathbf{b}) = -\frac{\mathbf{a}}{2} + 3\mathbf{b}$	M1A1 [4]
(b)	$\overrightarrow{OX} = \mu\overrightarrow{OB} = \mu 4\mathbf{b}$	M1
	$\overrightarrow{OX} = \overrightarrow{OM} + \overrightarrow{MX} = \overrightarrow{OM} + \lambda\overrightarrow{MY} = \mathbf{a} + \lambda\left(-\frac{\mathbf{a}}{2} + 3\mathbf{b}\right) = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\lambda\mathbf{b}$	M1
	$\Rightarrow \mu 4\mathbf{b} = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\mathbf{b}$	dM1
	$\Rightarrow 1 - \frac{\lambda}{2} = 0 \Rightarrow \lambda = 2$	
	$\Rightarrow 4\mu = 3\lambda \Rightarrow \mu = \frac{6}{4} = \frac{3}{2}$	ddM1
	$OB : OX = 2:3$ oe	A1 [5]
	ALT – working with alternative vector within triangle *OMX*	
	$\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OB} + \mu\overrightarrow{OB} = -\boldsymbol{a} + 4\boldsymbol{b} + \mu 4\boldsymbol{b}$	[M1
	$\overrightarrow{MX} = \lambda\left(-\frac{\boldsymbol{a}}{2} + 3\boldsymbol{b}\right)$	M1
	$\Rightarrow -\boldsymbol{a} = -\frac{\lambda\boldsymbol{a}}{2} \Rightarrow \lambda = 2$	dM1
	$\Rightarrow 4\boldsymbol{b} + \mu 4\boldsymbol{b} = 3\lambda\mathbf{b} \Rightarrow \mu = \frac{1}{2}$	ddM1
	$OB : OX = 2:3$ oe	A1]
(c)	$\frac{\Delta YBX}{\Delta ABX} = \frac{1}{4}$	M1
	$\frac{\Delta ABX}{\Delta OAX} = \frac{1}{3}$	
	$\Rightarrow \frac{\Delta YBX}{\Delta OAX} = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12} \Rightarrow \Delta YBX : \Delta OAX = 1:12$	M1A1 [3]
	ALT – working with relative areas of triangles	
	Area $\Delta YBX = a$ Area $\Delta ABX = 4a$ Area $\Delta OMX = 6a$ Area $\Delta AYX = 3a$ Area $\Delta OAY = 6a$ Area $\Delta OYB = 2a$	[M1
	Area $\Delta OAX = 12a$ $\Delta YBX : \Delta OAX = 1:12$	M1 A1]
		Total 11 marks

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Part	Mark	Notes
(a)	B1	For the correct simplified vector $\overrightarrow{AB}$
	M1	For the correct vector statement for $\overrightarrow{MY}$
	A1	For the correct simplified vector for $\overrightarrow{MY}$
(b)	M1	For the statement $\overrightarrow{OX} = \mu 4\mathbf{b}$ Note: this is a B mark on epen.
	M1	For the correct vector for $\overrightarrow{OX}$ (ft their MY)
	dM1	For equating both vectors for $\overrightarrow{OX}$ and for comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5
	ALT – working with alternative vector within triangle OMX	
	M1	For a vector statement which includes $\mu 4\boldsymbol{b}$ (for OX or BX) (ft their MY) Note: this is a B mark on epen.
	M1	For a correct second vector equation for the same vector (ft their MY)
	dM1	For equating both vectors and comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5 Condone $OX:OB = 3:2$ if clearly stated, but not just 3:2
(c)	M1	For either the relationship between the of areas of triangles BYX and ABX **or** the relationship between the areas of triangles ABX and OAX
	M1	For finding the relationship between the of areas of triangles BYX and OAX
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).
	ALT – working with relative areas	
	M1	For assigning a value to one triangle area and writing a second area in terms of this. Note: This could also follow from working with area of a triangle $= \frac{1}{2}ab\sin C$

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		e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	M1	For finding the relationship between the of areas of triangles BYX and OAX Note: This could also follow from working with area of a triangle $= \frac{1}{2} ab \sin C$ e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).

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Answer 6 of 8 · Question 10 part c · 4 marks

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10 a (i)	$\overrightarrow{OP} = \overrightarrow{OA} + \frac{3}{4}\overrightarrow{AB} = \mathbf{a} + \frac{3}{4}(-\mathbf{a}+\mathbf{b})$ or $\overrightarrow{OB} + \frac{1}{4}\overrightarrow{BA} = \mathbf{b} + \frac{1}{4}(\mathbf{a}-\mathbf{b})$	M1
	$\overrightarrow{OP} = \frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$	A1
(ii)	$\overrightarrow{MN} = \overrightarrow{MO} + \frac{1}{2}\overrightarrow{OP} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$ or $\frac{1}{2}\overrightarrow{AO} + \frac{1}{2}\overrightarrow{OP}$	M1
	$\overrightarrow{MN} = -\frac{3}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$	A1 (4)
b	$\overrightarrow{OC} = \lambda\mathbf{b}$	B1
	$\overrightarrow{AN} = \left(\overrightarrow{AO} + \overrightarrow{ON}\right) = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right) = -\frac{7}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or $\overrightarrow{AN} = \overrightarrow{AP} + \overrightarrow{PN} =$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \mu\overrightarrow{AN} = \mathbf{a} + \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$	M1 (A1 on ePen)
	$= \mathbf{a} - \frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ *or* $\left(1 - \frac{7}{8}\mu\right)\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$	A1 (M1 on ePen)
	$\therefore \lambda = \frac{3}{8}\mu$ and $0 = 1 - \frac{7}{8}\mu \Rightarrow \mu = \frac{8}{7}$ $\therefore \lambda = \frac{3}{7}$	M1 (A1 on ePen)
	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$	A1 (6)
ALT	$\overrightarrow{AC} = -\mathbf{a} + \lambda\mathbf{b}$	B1
	$\overrightarrow{AN} = \left(\overrightarrow{AO} + \overrightarrow{ON}\right) = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right) = -\frac{7}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or $\overrightarrow{AN} = \overrightarrow{AP} + \overrightarrow{PN} =$	M1
	$\overrightarrow{AC} = \mu\overrightarrow{AN} = \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$	M1 (A1 on ePen)
	$= -\frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$	A1 (M1 on ePen)
	$\therefore \lambda = \frac{3}{8}\mu$ and $0 = 1 - \frac{7}{8}\mu \Rightarrow \mu = \frac{8}{7}$ $\therefore \lambda = \frac{3}{7}$	M1 (A1 on ePen)
	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$	A1 (6)

General principles for marking part (b) – if in any doubt about allocating marks – send to review

B1 Writes a valid vector with a parameter in terms of **a** or **b** which leads to finding $\overrightarrow{OC}$

M1 M1 A1 Writes a second valid vector with a different parameter, in terms of **a** and **b**, following a distinct different route, which leads to finding $\overrightarrow{OC}$

M1 Compares components with two different parameters and arrives at a value for μ or λ

A1 correct vector

Type MS · Using vector ratios to compare areas

Answer 6 of 8 · Question 10 part c · 4 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

c	(Area triangle) $OAP = \frac{3}{4}$ (Area triangle) OAB	B1
	(Area triangle) $OMN = \frac{1}{4}$ (Area triangle) $OAP = \frac{3}{16}$ (Area triangle) OAB	B1
	(Area quadrilateral) $AMNP =$ $\frac{3}{4}$ (Area triangle) $OAB - \frac{3}{16}$ (Area triangle) OAB	M1
	$= \frac{9}{16}$ (Area triangle) OAB $\quad k = \frac{9}{16}$	A1 (4)
ALT	$\frac{(\text{Area triangle})OAP}{(\text{Area triangle})OAB} = \frac{3}{4}$	B1
	$\left(\frac{\text{Area triangle } OMN}{\text{Area triangle } OAP} = \frac{1}{4}\right) \Rightarrow \frac{(\text{Area quadrilateral})MNAP}{(\text{Area triangle})OAP} = \frac{3}{4}$	B1
	$\frac{(\text{Area quadrilateral})MNAP}{(\text{Area triangle})OAP} \times \frac{(\text{Area triangle})OAP}{(\text{Area triangle})OAB} = \frac{3}{4} \times \frac{3}{4}$	M1
	$k = \frac{9}{16}$	A1 (4)
		Total 14 marks

Type MS · Using vector ratios to compare areas

Answer 6 of 8 · Question 10 part c · 4 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Part	Marks	Note
(a) (i)	M1	For stating **or** using $\overrightarrow{OP} = \overrightarrow{OA} + \frac{3}{4}\overrightarrow{AB}$ or for $\mathbf{a} + \frac{3}{4}(-\mathbf{a} + \mathbf{b})$ or $\overrightarrow{OB} + \frac{1}{4}\overrightarrow{BA}$ or $\mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ (can be implied by correct vector)
	A1	For $\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ or valid alternative form such as $\frac{\mathbf{a}+3\mathbf{b}}{4}$ or $\frac{1}{4}(\mathbf{a}+3\mathbf{b})$
(ii)	M1	For stating **or** using $\overrightarrow{MN} = \overrightarrow{MO} + \frac{1}{2}\overrightarrow{OP}$ or $\frac{1}{2}\overrightarrow{AO} + \frac{1}{2}\overrightarrow{OP}$ or $-\frac{1}{2}\mathbf{a} + \frac{1}{2}\left(\text{their } \overrightarrow{OP}\right)$
	A1	$-\frac{3}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or valid alternative form such as $\frac{-3\mathbf{a}+3\mathbf{b}}{8}$ or $\frac{1}{8}(-3\mathbf{a}+3\mathbf{b})$ (can be implied by correct vector)
(b)	B1	$\overrightarrow{OC} = \lambda\mathbf{b}$ or any equivalent statement involving a parameter, in terms of **b**.
	M1	A fully correct method to find $\overrightarrow{AN}$ using their $\overrightarrow{OP}$ $\overrightarrow{AN} = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$
	M1 (A1 on ePen)	**Using** $\overrightarrow{OC} = \overrightarrow{OA} + \mu\left(\text{their } \overrightarrow{AN}\right)$ in terms **a** and **b** $\overrightarrow{OC} = \overrightarrow{OA} + \mu\overrightarrow{AN} = \mathbf{a} + \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$ Simplification not required
	A1 (M1 on ePen)	$= \mathbf{a} - \frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ *or* $\left(1 - \frac{7}{8}\mu\right)\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ either of these forms, ready for comparing coefficients
	M1 (A1 on ePen)	For correctly equating their components with two different parameters and attempting to solve, reaching values for μ or λ We need to see two equations here, leading to a value for one of the parameters.
	A1	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$
ALT	B1	$\overrightarrow{AC} = -\mathbf{a} + \lambda\mathbf{b}$ or any equivalent statement involving a parameter, in terms of **a** and **b**.
	M1	A fully correct method to find $\overrightarrow{AN}$ using their $\overrightarrow{OP}$ $\overrightarrow{AN} = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$
	M1 (A1 on ePen)	$\overrightarrow{AC} = \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$ using their $\overrightarrow{AN}$
	A1 (M1 on ePen)	Correct vector for $\overrightarrow{AC}$
	M1 (A1 on ePen) A1	Marks allocated as main scheme
(c) BOTH SCHEMES	B1	Correct statement
	B1	Correct statement
	M1	Uses their statements to carry out a relevant calculation
	A1	Correct value for k

Type MS · Using vector ratios to compare areas

Answer 6 of 8 · Question 10 part c · 4 marks

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Type MS · Using vector ratios to compare areas

Answer 7 of 8 · Question 9 part c.ii · 4 marks

2025 NOV · Paper 02 · extract 1 of 5 · find part c.ii in this mark scheme extract

	Scheme	Marks
9(a)	$\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$ or $-(2\mathbf{a}+4\mathbf{b})+(7\mathbf{a}+7\mathbf{b})$	B1
	$= 5\mathbf{a}+3\mathbf{b}$	B1
	$\overrightarrow{BC}\left[= \frac{1}{2}(10\mathbf{a}+6\mathbf{b})=\right] \frac{1}{2}\overrightarrow{AD}$ oe eg $\frac{1}{2}(10\mathbf{a}+6\mathbf{b}) = 5\mathbf{a}+3\mathbf{b}$ eg $10\mathbf{a}+6\mathbf{b} = 2(5\mathbf{a}+3\mathbf{b})$	M1
	[therefore parallel] therefore trapezium or eg # or shown.	A1 [4]
(b)	$\left[\overrightarrow{BD} = -\overrightarrow{AB} + \overrightarrow{AD} = -(2\mathbf{a}+4\mathbf{b})+(10\mathbf{a}+6\mathbf{b}) =\right] 8\mathbf{a}+2\mathbf{b}$	B1
	$\left[\lvert BD\rvert =\right] \sqrt{8^2+2^2}\left[=\sqrt{68}=2\sqrt{17}\right]$	B1ft
	Unit vector is $\pm\frac{1}{\sqrt{68}}(8\mathbf{a}+2\mathbf{b})$ oe eg $\pm\frac{1}{\sqrt{17}}(4\mathbf{a}+\mathbf{b})$	M1A1 [4]

Type MS · Using vector ratios to compare areas

Answer 7 of 8 · Question 9 part c.ii · 4 marks

2025 NOV · Paper 02 · extract 2 of 5 · find part c.ii in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct unsimplified vector in **a** and **b** for $\overrightarrow{BC}$ or for $\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$
	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BC}$
	M1	For $\overrightarrow{BC} = \frac{1}{2}\overrightarrow{AD}$ oe (note accept eg $2BC = AD$ ie without vector notation for this mark) Accept equivalences – see example MS.
	A1	For the correct conclusion, with all previous marks gained. Ignore work on other vectors. Do not accept eg $2BC = AD$ without vector notation for this mark Note B0 B1 M1 A0 is a potential score. $2BC = AD$ followed by eg $10\mathbf{a} + 6\mathbf{b} = 10\mathbf{a} + 6\mathbf{b}$ will get Aq
Watch for equivalences such as working out vector $\overrightarrow{CB}$ and stating $\overrightarrow{CB} = -\frac{1}{2}\overrightarrow{AD}$ Permit eg $\frac{\overrightarrow{AD}}{\overrightarrow{BC}} = 2$		
(b)	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BD}$
	B1ft	For applying Pythagoras theorem fully and correctly to find the length BD Ft any vector believed to be their $\overrightarrow{BD}$. $8^2 + 2^2$ alone is insufficient.
	M1	For using their vector for $\overrightarrow{BD}$ with their length BD to attempt to form a unit vector
	A1	For the correct unit vector.

Type MS · Using vector ratios to compare areas

Answer 7 of 8 · Question 9 part c.ii · 4 marks

2025 NOV · Paper 02 · extract 3 of 5 · find part c.ii in this mark scheme extract

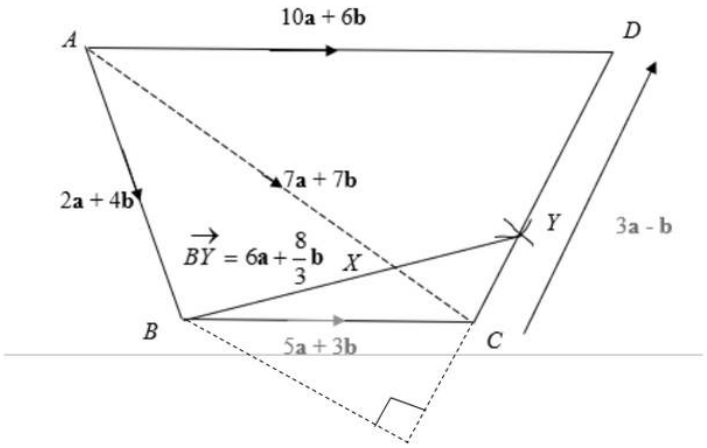
(c)(i)	$\left[\left[\overrightarrow{CY}=\right]\frac{1}{3}(3\mathbf{a}-\mathbf{b})=\mathbf{a}-\frac{1}{3}\mathbf{b}$ oe eg $\overrightarrow{YC}$ or $\left[\left[\overrightarrow{YD}=\right]\frac{2}{3}(3\mathbf{a}-\mathbf{b})=2\mathbf{a}-\frac{2}{3}\mathbf{b}$ oe eg $\overrightarrow{DY}$ Note: $\overrightarrow{BC}=5\mathbf{a}+4\mathbf{b}$ $\left[\overrightarrow{BY}=5\mathbf{a}+3\mathbf{b}+\mathbf{a}-\frac{\mathbf{b}}{3}=\right]6\mathbf{a}+\frac{8}{3}\mathbf{b}\Rightarrow$ $\left[\overrightarrow{BX}=\right]\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{BX}=\right]-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $\left[\overrightarrow{BX}=\right](5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})$ $\left[\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $(5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})\right]$ $\left[6\lambda=7\mu-2$ and $\frac{8}{3}\lambda=7\mu-4$ or $\frac{8}{3}\lambda=-7\alpha+3$ and $6\lambda=-7\alpha+5\right]$ $\left[\lambda=\frac{3}{5};\mu=\frac{4}{5};\alpha=\frac{1}{5}\Rightarrow\overrightarrow{BX}=\right]\frac{18}{5}\mathbf{a}+\frac{8}{5}\mathbf{b}\Rightarrow[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]
ALT	First mark as main scheme. $\left[\overrightarrow{XY}=\right]\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{XY}=\right]\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\left[\overrightarrow{XY}=\right]\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)$ $\left[\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)\right]$ $\left[6\beta=7\gamma+1$ and $\frac{8}{3}\beta=7\gamma-\frac{1}{3}$ or $\frac{8}{3}\beta=-7\delta+6+\frac{2}{3}$ and $6\beta=-7\delta+10-2\right]$ $\left[\beta=\frac{2}{5};\gamma=\frac{1}{5};\delta=\frac{4}{5}\Rightarrow\overrightarrow{XY}=\right]\frac{12}{5}\mathbf{a}+\frac{16}{15}\mathbf{b}[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]

Type MS · Using vector ratios to compare areas

Answer 7 of 8 · Question 9 part c.ii · 4 marks

2025 NOV · Paper 02 · extract 4 of 5 · find part c.ii in this mark scheme extract

Part	Mark	Notes
(c)(i) Main and ALT	M1	Correct simplified vector in **a** and **b** for $\pm \overrightarrow{BY}$
	dM1	Correct vector in **a** and **b** with parameter using $\overrightarrow{CY}$ or $\overrightarrow{YC}$ for $\overrightarrow{BX}$ / $\overrightarrow{XB}$ or $\overrightarrow{XY}$ / $\overrightarrow{YX}$ **and** correct vector in **a** and **b** with a different parameter along a second distinct path using $\overrightarrow{AC}$ or $\overrightarrow{CA}$ for the same vector. Dependent on previous method mark.
	ddM1	For the correct simultaneous equations. Any one correct parameter or $\overrightarrow{BX} = \frac{18}{5}\mathbf{a} + \frac{8}{5}\mathbf{b}$ will imply this mark. Dependent on both previous method marks.
	A1	For the required ratio $BX : XY = 3 : 2$
Note, correct method can include the opposite signs in vectors multiplied by a parameter, as the parameter will just be the opposite sign when calculated and a fully correct solution is possible. Eg $\left[\overrightarrow{BX} = \right](5\mathbf{a}+3\mathbf{b}) + \alpha(7\mathbf{a}+7\mathbf{b})$ would just make $\alpha = -\frac{1}{5}$ Vectors questions notoriously have different methods. Correct answer, not from obviously incorrect working, full marks may be awarded **if method is shown.** (This is because the demand of this question says students must use a vector method). Min method must involve finding two correct different distinct relevant paths.		



Type MS · Using vector ratios to compare areas

Answer 7 of 8 · Question 9 part c.ii · 4 marks

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(ii)	$\left[\frac{CXY}{BXC}=\frac{2}{3}\text{ and }\frac{BAC}{BXC}=\frac{5}{1}\Rightarrow\right]\ \frac{CXY}{BAC}=\frac{2}{15}$ oe eg $\frac{BAC}{CXY}=\frac{15}{2}$	M1
Eg 1	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{CXY}{ABCD}=\frac{\frac{2}{15}}{3}\left[=\frac{2}{45}\right]$ oe eg $\frac{ABCD}{CXY}=\frac{3}{\frac{2}{15}}\left[=\frac{45}{2}\right]$	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
Eg 2	$\frac{CXY}{BCY}=\frac{2}{5}$ and $\left[\frac{BCY}{BCD}=\frac{1}{3}\right]\ \frac{BCY}{BCA}=\frac{1}{3}$ oe	M1
	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{BCY}{ABCD}=\frac{\frac{1}{3}}{3}\left[=\frac{1}{9}\right]$ oe and $\frac{CXY}{ABCD}=\frac{\frac{2}{5}}{9}\left[=\frac{2}{45}\right]$ oe	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
		Total 16 marks

Type MS · Using vector ratios to compare areas

Answer 8 of 8 · Question 8 part e · 4 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

8 (a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \mathbf{a} + \mathbf{b}$	B1 [1]
(b)	$\overrightarrow{OC} = 2\mathbf{b}$ $\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b} = \mathbf{b} - \mathbf{a}$	B1 M1A1 [3]
(c)	$\overrightarrow{OM} = \overrightarrow{OB} + \overrightarrow{BM} = \mathbf{a} + \mathbf{b} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}$ or $\frac{1}{3}(\mathbf{a} + 5\mathbf{b})$	 M1A1ft [2]
(d)	$\overrightarrow{OY} = \mu\left(\frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}\right) = \frac{\mu\mathbf{a}}{3} + \frac{5\mu\mathbf{b}}{3}$ $\overrightarrow{OY} = \overrightarrow{OA} + \overrightarrow{AY} = \boldsymbol{a} + \lambda\boldsymbol{b}$ $\Rightarrow \frac{\mu}{3} = 1$ and $\frac{5\mu}{3} = \lambda$ Solves simultaneous equations by any method $\mu = 3,\ \lambda = 5$ $AB : BY = 1 : 4$ **ALT** $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ $\Rightarrow -1 + \frac{1}{3}\mu = 0$ and $\lambda = -1 + \frac{5}{3}\mu$ $\mu = 3, \lambda = 4$ $AB : BY = 1 : 4$	M1 M1 M1 M1 A1 [5] [M1 M1 M1 M1 A1]
(e)	$10 = \frac{1}{2}ab\sin 60^\circ \Rightarrow ab = \frac{40}{\sqrt{3}} \Rightarrow a = \frac{40}{b\sqrt{3}}$ $\text{Area} = \frac{1}{2}a \times 5b\sin 120^\circ = \frac{1}{2} \times \frac{40}{b\sqrt{3}} \times 5b\sin 120^\circ$ $\text{Area} = 50$ **ALT** $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2} \times h \times 5}{\frac{1}{2} \times h \times 1} = \frac{5}{1}$ $\text{Area } OAY = 5 \times \text{Area } OAB$ $\text{Area } OAB = 10$ $\text{Area} = 50$	M1A1 dM1 A1 [4] [M1 A1 dM1 A1]
		Total 15 marks

Type MS · Using vector ratios to compare areas

Answer 8 of 8 · Question 8 part e · 4 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

Part	Mark	Notes
(a)	B1	For a correct expression for $\overrightarrow{OB}$ in terms of **a** and **b**
(b)	B1	For a correct expression for $\overrightarrow{OC}$
	M1	For a correct vector statement for $\overrightarrow{BC}$: $\overrightarrow{BC}=\overrightarrow{BO}+\overrightarrow{OC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$. Vector statement must be suitable for substitution to find $\overrightarrow{BC}$
	A1	For the correct simplified $\overrightarrow{BC}$ in terms of a single **a** and **b** only. $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ If answer $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ seen without wrong working then award B1M1A1.
(c)	M1	For a correct vector statement for $\overrightarrow{OM}$: $\overrightarrow{OM}=\overrightarrow{OB}+\frac{2}{3}\overrightarrow{BC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$ and their $\overrightarrow{BC}$.
	A1ft	For the correct simplified using their $\overrightarrow{OM}$ in terms of a single **a** and **b** only. $\overrightarrow{OM} = \frac{\boldsymbol{a}}{3} + \frac{5\boldsymbol{b}}{3}$ or $\frac{1}{3}(\boldsymbol{a} + 5\boldsymbol{b})$
(d)	M1 M1	M1 for one correct statement of route for $\overrightarrow{OY}$ M1 for second correct statement of route for $\overrightarrow{OY}$ $\overrightarrow{OY} = \mu\overrightarrow{OM}$ (or any other variable in place of μ) $\overrightarrow{OY} = \left(\overrightarrow{OA} + \overrightarrow{AY}\right) = \mathbf{a} + \lambda\mathbf{b}$ (or any other variable in place of λ, provided this is different to their μ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$
ALT – use of $\overrightarrow{BY}$ route, use also for $\overrightarrow{AY}$ route		
	M1 M1	M1 for one correct statement of route for $\overrightarrow{BY}$ M1 for second correct statement of route for $\overrightarrow{BY}$ $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ (or any other variable in place of λ) $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ (or any other variable in place of μ provided this is different to their λ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$

Type MS · Using vector ratios to compare areas

Answer 8 of 8 · Question 8 part e · 4 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract

(e)	**M1**	For use of the **correct** formula for area of a triangle with $60°$ and correct value of $\sin 60°$ $10 = \frac{1}{2}ab\sin 60° = ab \times \frac{\sqrt{3}}{4}$
	A1	For correct expression for ab or a $ab = \frac{40}{\sqrt{3}}$ or $a = \frac{40}{b\sqrt{3}}$
	dM1	For use of the **correct** formula for area of a triangle with $120°$ and attempt to substitute for ab or a. $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times '\frac{40}{b\sqrt{3}}' \times '5'b\sin 120°$ or $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times ab \times '5'\sin 120° = \frac{1}{2} \times '\frac{40}{\sqrt{3}}' \times '5'\sin 120°$ Dependent on the first M awarded. Allow use of their λ from part (d).
	A1	For the correct area Area $= 50$
ALT – use of ratios of areas		
	M1	For use of their ratio $AB:BY$ to write an equation linking area OAY and area OAB $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2}\times h\times(1+'4')}{\frac{1}{2}\times h\times 1} = \frac{'5'}{1}$
	A1	For correct relationship between area OAY and area OAB
	dM1	For a correct method to find the area of OAB Dependent on first M mark being awarded.
	A1	For the correct area Area $= 50$

Type MS · Using vector ratios to compare areas

Answer 8 of 8 · Question 8 part e · 4 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

Question number	Scheme	Marks